RESEARCH ROADMAP

A Graph Reinforcement Learning Framework for Routing in a Hybrid Fiber and Free-Space QKD Network

Agent-executable research, experiment, validation, and paper-writing plan

Visual system matched to the uploaded QSMS Presentation Script: white background, DejaVu Sans typography, burgundy #9E1735 headings/lines, pale burgundy #F6F1F2 panels, dark #242424 body text.

This document is intentionally operational. It tells a research agent what is already implemented, what evidence exists, what is still missing, which dependency unlocks each task, where each equation/reference belongs, and what figure/table should be produced from each experiment.

Source set

1. Current draft: QKD_IEEE_Paper.pdf. 2. Base paper: Johann et al., JOCN 2024, Comparison and optimization of different routing methods for meshed QKD networks using trusted nodes. 3. Uploaded QSMS script used only as the visual/font/color reference for this roadmap.

1. STATUS LEGEND — HOW AN AGENT SHOULD READ THIS

STATUS	Meaning	Agent action
DONE	Implemented and described in the current draft, or supported by an archived result.	Do not rebuild from scratch. Verify reproducibility and improve presentation/validation if marked elsewhere.
PARTIAL	Implementation exists but evidence is incomplete, single-condition, assumption-heavy, or not independently validated.	Treat as existing infrastructure; prioritize validation before making comparative claims.
ARCHIVED	A result/configuration exists, but it is not sufficient as final evidence.	Preserve it as provenance. Regenerate under the corrected protocol before using it as a main result.
TODO	Not implemented or not measured.	Build only after prerequisites are satisfied.
BLOCKED	Cannot be meaningfully completed until another model/data layer exists.	Do not fake a result. Complete the dependency first.

Non-negotiable rule: a simulator quantity is not a measurement simply because it has a numerical value. The current paper explicitly distinguishes modeled outputs, assumptions, calibration evidence, and field validation.

2. CURRENT STATE — WHAT IS ALREADY DONE

Workstream	Status	Evidence / current implementation	Final-paper use
Hybrid topology	**DONE**	7 Indian city hubs; 2,370 modeled nodes; 2,653 links; 273 fiber + 2,380 candidate FSO. Road geometry is a routing proxy.	Methods + topology figure; explicitly label as proxy.
Fiber model	**PARTIAL**	1550 nm; 0.20 dB/km; detector efficiency 0.93; dark-count bound <70 s^{-1}; excess loss included.	Methods + analytical validation; provenance table.
FSO model	**PARTIAL**	785 nm scenario; log-normal correlated turbulence; atmospheric/diffraction/pointing/background terms; detector noise. No Gamma-Gamma or generalized amplitude damping.	Methods + stress experiments; assumptions must remain visible.
Rate model	**PARTIAL**	r_proxy = p_det[1−2h_2(Q)]; normalized simulator utility, not bps and not a composable security certificate.	Methods Eq. (1) + explicit scope limitation.
QBER feasibility	**DONE**	Hard Q_max = 0.11 action mask; projected chain-error constraint also used.	Methods + correctness tests.
Observation	**DONE**	5 node features + 9 edge features, including QBER, rate proxy, FSO flag, distance, projected error, pool state, switching, visited flag, progress.	Methods Table II + ablation.
GNN-PPO	**DONE**	3 message-passing layers; shared actor-critic trunk; masked action head; PPO; optional attention.	Methods + architecture figure + ablation.
Classical baselines	**PARTIAL**	Random feasible, BFS, Dijkstra, Max-SKR/rate selector exist; old baseline outputs invalid because selector did not consistently honor mask.	Regenerate all under one corrected protocol.
Archived GNN result	**ARCHIVED**	Mumbai→Kolkata, monsoon, 22:00±1 h, 200 epochs resumed; 10/10 success; mean 25.6 hops.	Use only as preliminary/single-condition evidence.
Shortest-path reference	**DONE**	Independent NetworkX check: 24-edge unweighted shortest path; old 23-edge reference was incorrect.	Correct the paper and all plots/captions.
Controlled link-choice diagnostic	**ARCHIVED**	384 paired comparisons; FSO higher proxy in all sampled pairs; trained policy selected FSO zero times; action probability near 1/2.	Main diagnostic showing current policy weakness; motivate ablations.
Physics evidence	**PARTIAL**	80-km fiber: 16.46 dB modeled loss, p_det≈0.021, QBER≈0.0105, proxy≈0.0175. 10-km FSO: modeled availability 88% config / 88.5% in 1,000 samples; median conditional proxy 0.2035.	Report as modeled outputs, not measured performance.
ILP	**PARTIAL**	Mathematical comparator Eq. (5)–(8) already written in current draft; not executable/measured because demand/QKP units are absent.	Keep as future/global reference until demand layer exists.

Current evidence is therefore strongest at the simulator + route-controller level, not yet at the sustained network-service level.

3. WHAT IS STILL MISSING — IN DEPENDENCY ORDER

Priority	Missing layer	Dependency	Why it matters	Paper consequence
P0	Corrected baseline regeneration	Gymnasium/runtime + common mask	All comparative claims currently depend on this.	Replace old baseline table; add statistically valid comparison.
P0	5-seed training/evaluation	Stable evaluation runtime	One archived run cannot establish policy behavior.	Report mean/spread and seed sensitivity.
P0	Physics/noise analytical validation	Independent calculations / code checks	Separates model correctness from RL behavior.	Strengthens Methods + Validation.
P0	Controlled link-choice/QBER/FSO stress	Existing simulator	Tests whether the policy actually uses physical features.	Core mechanism result.
P1	Endpoint-pair generalization	Multiple endpoint runs	Current result is one pair.	Generalization subsection.
P1	Season/diurnal generalization	Scenario generator	Current result is one monsoon-night condition.	Robustness subsection.
P1	Feature/architecture/reward ablations	Training pipeline	Current policy weakness needs attribution.	Ablation table/figures.
P1	Inference benchmark	Runtime harness	Current paper has no measured target-hardware latency.	Deployment/complexity subsection.
P1	Exact small-graph oracle	Tiny graph generator	Provides route-quality reference without ILP scope explosion.	Oracle/regret result.
P2	Absolute QKP/key resource model	Absolute key-rate model	Normalized pool cannot support network-level service metrics.	Unlocks Johann-style experiments.
P2	Demand matrix + temporal demand	Trace or declared synthetic model	Needed for blocking/load balancing/prediction.	Unlocks network-service study.
P2	Key consumption + refill	Absolute rate + QKP capacity	Creates temporal resource coupling.	Network-operation simulator.
P2	Blocking/load balancing	Demand + QKP loop	Directly comparable to Johann KPIs.	Journal-level result.
P2	Management traffic	Controller/query protocol definition	Cannot measure overhead without message semantics.	Security/overhead result.
P2	Weight optimization	QKP state + demand loop	Classical state-aware comparator.	Journal baseline.
P2	LSTM demand prediction	Historical demand matrices	Only meaningful with dynamic demand.	Journal temporal comparator.

Priority	Missing layer	Dependency	Why it matters	Paper consequence
P2	Perfect prediction	Future demand matrix	Oracle upper reference for prediction methods.	Journal reference.
P2	ILP execution	Demand matrix + absolute capacity + candidate paths	Current ILP is only a formulation.	Global demand-level reference.
P3	Measured traces / HIL / field	Calibrated hardware/channel data	External validation only after simulator is stable.	Validation/limitations.

4. P0 — IMMEDIATE PAPER EXPERIMENTS

ID	Experiment	Procedure	Output	Write it as
P0-1	Corrected baselines	Run Random-feasible, BFS, Dijkstra, Max-SKR with identical state, endpoint, seed set, termination rule and valid-action mask.	CSV + summary table + distributions.	“Under a common feasibility protocol…”
P0-2	Physics checks	Sweep fiber distance; independently calculate loss/detection; sweep QBER/noise; validate rate-proxy monotonicity; sample FSO availability/correlation.	4–6 validation plots.	“The simulator reproduces the expected limiting trends…”
P0-3	Link-choice sensitivity	Construct matched candidate links with controlled ΔR, QBER margin, distance and outage; hold route progress constant; record action probability and selected action.	Probability-vs-ΔR scatter/curve + selection-rate plot.	“The policy does/does not respond monotonically to…”
P0-4	QBER stress	Evaluate QBER levels 0.01, .03, .05, .07, .09, .10, .109 and >.11; separate hard-mask behavior from preference among feasible links.	Success/entropy/probability curves.	“Feasibility and preference are evaluated separately…”
P0-5	FSO stress	Vary turbulence, availability, pointing/background and outage persistence independently where supported.	Robustness curves/heatmap.	“Performance degrades/recoveries under controlled channel perturbations…”
P0-6	5-seed stability	Train 5 independent policies with identical protected hyperparameters; fixed evaluation seeds.	Return/success/hops distributions + training curves.	“Across independent training seeds…”

Stop condition for the immediate paper: do not expand to demand prediction or ILP experiments until P0 produces a trustworthy baseline comparison and establishes whether the learned policy actually uses the physical link features.

5. P1 — GENERALIZATION, ABLATION, AND INFERENCE

ID	Work	Minimum design	Primary figure/table
P1-1	Endpoint generalization	Train on selected pairs; test held-out source–destination pairs among the 7 hubs.	Train/test matrix or held-out endpoint bar/box plot.
P1-2	Season/diurnal	Clear/monsoon × day/night; hold out at least one condition from training.	Condition heatmap: success, hops, proxy, outage.
P1-3	Message-passing ablation	0/1/2/3 layers under matched parameter/training budget where possible.	Ablation bar/box plot.
P1-4	Attention ablation	Same seed set and budget, GNN vs attention.	Paired distribution + action-probability diagnostic.
P1-5	Feature ablation	Remove QBER, rate, pool, progress, channel-type features one group at a time.	Performance delta table + tornado-style bar chart.
P1-6	Reward ablation	Remove/perturb rate, pool, delay, energy, congestion, switching, progress terms.	Metric matrix; avoid claiming causality from one run.
P1-7	Small-graph oracle	Enumerate feasible paths on tiny graphs and compare selected route utility/regret.	Regret CDF / route-quality scatter.
P1-8	Inference benchmark	Measure model forward, decision, route, p50/p95/p99, CPU/GPU, memory; scale graph size.	Latency distribution + latency-vs-graph-size plot.

Inference comparison rule: use Dijkstra as the primary classical per-decision reference and Max-SKR as a second reference. Do not put ILP on the same per-next-hop latency plot; ILP is a global demand-level optimizer.

6. P2 — JOURNAL EXTENSION: THE NETWORK-SERVICE LAYER

Target transformation

Current paper: channel state → route → next hop. Journal extension: demand → route → key consumption → QKP depletion → QKD generation/refill → next demand.

Stage	Implement	Then measure	Unlocks
J-1 Absolute key resources	Per-edge key capacity, initial key count, key bits/keys, absolute generation rate.	QKP inventory over time.	Resource-aware routing.
J-2 Demand stream	Time-indexed source/destination/data requests; preserve trace provenance.	Demand volume and temporal variation.	Blocking/load balancing.
J-3 Consumption	For a selected path, consume required keys from every QKP.	Per-demand accepted/rejected state.	Finite resource coupling.
J-4 Refill	Generate/distribute keys according to link generation rate at defined intervals.	QKP recovery curves.	Sustained operation.
J-5 Blocking	Reject demand if any path QKP lacks sufficient keys.	Blocking probability vs traffic factor.	Direct Johann KPI.
J-6 Load balance	Track mean/min/variance/filling levels per edge/region.	QKP utilization heatmaps/time series.	Traffic-engineering result.
J-7 Management traffic	Define controller→QKP queries and count them per routing demand/time interval.	Requests/bytes vs traffic.	Information-overhead result.

Base-paper provenance: Johann et al. describe the centralized controller/QKP architecture on p. 385, the dynamic demand/key-consumption/refill loop on p. 386, blocking on p. 389, QKP utilization/load balancing on p. 389–390, and management traffic on p. 390. Their study uses 10,000 initial keys of 256 bits per QKP and periodic generation/refill; if reproducing those assumptions, cite Johann directly rather than presenting them as your own model.

7. EQUATION REGISTRY — EXACTLY WHERE TO GET EACH EQUATION

Equation / object	Use in paper	Exact source	Agent instruction
Normalized rate proxy	Current link-utility definition.	Current draft, Section II-B, Eq. (1), PDF p. 4.	Copy the existing equation exactly and label it a simulator proxy. Do not call it bps or a composable key rate.
Discounted RL objective J(θ)	Formal routing objective.	Current draft, Section IV-A, Eq. (2), PDF p. 5; RL foundation cited to [8], [25], [26].	Retain current notation. Explain hard feasibility separately from reward optimization.
Masked policy πθ(u|o)	Hard-feasibility action selection.	Current draft, Section IV-C-2, Eq. (3), PDF p. 5.	Use current equation. Explain that invalid actions are removed before softmax.
PPO clipped surrogate	Training objective.	Current draft, Section IV-C-3, Eq. (4), PDF p. 5; original PPO = reference [8].	Keep equation from current draft; cite Schulman et al. [8] for PPO.
GNN/message passing	Architecture definition.	Current draft Section IV-C-2; cite Kipf & Welling [9], Hamilton et al. [27], Gilmer et al. [28] depending on exact layer formulation.	Do not copy an external equation unless it is the actual layer used. If the implementation has a custom update, derive the equation from the code.
Attention	Optional architecture.	Current draft Section IV-C-2; cite Velickovic et al. [10].	Only write a mathematical attention equation if the implemented attention layer is explicitly characterized and ablated.
Fiber/FSO physical equations	Channel model.	Current draft states the implementation and points to PHYSICS_EVIDENCE.md; detailed code/model provenance is outside the paper text.	Extract the exact implemented equations from the physics evidence/code. Do not invent a textbook equation and claim it is the implementation.
Johann QBER model	Only if adopting Johann's absolute-rate model.	Johann paper p. 384, Eq. (1).	Use only if you actually port that model. Cite Johann and preserve parameter definitions.
Johann post-processing factor	Absolute key generation extension.	Johann p. 384, Eq. (2).	Use with the same caveat: only after adopting their model and assumptions.
Johann Nsec	Key-block generation extension.	Johann p. 384, Eq. (3).	Useful for converting absolute key-generation model into 256-bit key blocks if you reproduce that framework.
Weight heuristic We = exp(aEe)	Classical state-aware comparator.	Current draft Section IV-D-1, Eq./definition text; attributed to [4].	Implement only with QKP state. Cite Johann [4]. A rate-aware modification is a new method and must be labeled as such.
ILP objective/constraints	Global demand-level comparator.	Current draft Section V, Eqs. (5)–(8), PDF p. 6; formulation is mathematical only at present.	Do not report performance until demand/capacity simulation exists. Use as global reference, not next-hop baseline.

Equation provenance rule: every equation in the final paper should be traceable to one of three sources: (A) already implemented/current draft; (B) a cited external reference whose model is actually adopted; or (C) a new equation derived from the proposed method and supported by the implementation. Never insert an equation merely because it is standard if the simulator does something different.

8. REFERENCE MAP — WHAT TO CITE FOR WHAT

Ref.	Use it for	Do not use it for
[4] Johann et al., JOCN 2024	Trusted-node routing, QKP architecture, demand simulation, k-shortest paths, weight optimization, blocking, load balancing, management traffic.	Do not claim their absolute key-rate model is your implemented model unless ported.
[8] Schulman et al., PPO	PPO algorithm and clipped objective.	Not a QKD routing reference.
[9] Kipf & Welling, GCN	Graph convolution/message aggregation background.	Not necessarily the exact architecture unless your layer matches.
[10] Velickovic et al., GAT	Attention mechanism background.	Do not imply attention improved performance without your ablation.
[11] Bennett & Brassard	BB84 background.	Not a finite-key certificate for your hybrid network.
[12], [13] Hwang; Lo–Ma–Chen	Decoy-state background.	Do not claim the current proxy implements their full security analysis.
[14] Gottesman et al.	Security with imperfect devices.	Use as security background, not as validation of the simulator.
[15] Tomamichel et al.	Finite-key analysis foundations.	Use if discussing finite-size security limitations or future rigorous estimator.
[16] Lim et al.	Practical decoy-state security bounds.	Same boundary: background unless actually implemented.
[17] ITU-T Y.3803	QKD key-management/network architecture context.	Not a source of measured performance.
[18] ITU-R P.1814-1	Terrestrial FSO link-design/prediction context.	Not a calibration dataset for the current Indian topology.
[19] Krzic et al., npj Quantum Information 2023	Metropolitan free-space quantum network context.	Not direct validation of your weather assumptions.
[20] Dijkstra	Classical shortest-path baseline.	Not a QKD-specific resource-aware algorithm.
[21] Roik et al.	Related reinforcement-learning routing in quantum networks.	Use for positioning, not direct apples-to-apples comparison.
[22] Amer et al.	Dynamic QKD routing context.	Use for related-work positioning and future demand-aware extension.
[23] Bellman	Dynamic programming background if needed.	Not necessary for the core paper unless discussing classical sequential optimization.
[24] Schulman et al., GAE	Generalized advantage estimation.	Use only in RL methodology.

Ref.	Use it for	Do not use it for
[25], [26] POMDP/RL foundations	Partial observability and RL formulation.	Not evidence that the specific QKD environment is correct.
[27], [28], [29] GNN/MPNN/attention foundations	Graph representation and attention theory.	Use only where the implemented architecture corresponds.
[30] OSM	Map-data attribution.	Do not describe road corridors as verified fiber routes.

9. FIGURE PLAN — EXACTLY WHAT TO DRAW

Fig.	Priority	Figure	How to generate it	Message/caption
F1	KEEP/UPDATE	Normalized rate proxy vs link distance	Sweep fiber/FSO distance under fixed scenarios; shade only sampled simulator variation.	“Modeled normalized rate proxy; not measured link performance.”
F2	KEEP	TN vs STN conceptual architecture	Existing conceptual schematic; retain clear boundary between conventional TN relay and STN abstraction.	“Security architecture context; STN proof is not implemented.”
F3	KEEP/REFINE	Hybrid routing architecture	Draw topology → environment → GNN/PPO → action mask → route/key-pool state. Use current Fig. 3 concept.	Show what is implemented vs conceptual OTP/XOR/key-management plane.
F4	KEEP/UPDATE	Indian corridor graph	Plot 2,370 nodes/2,653 links; distinguish fiber and candidate FSO; mark Mumbai→Kolkata example.	Explicitly label road geometry/FSO visibility as proxies/assumptions.
F5	KEEP	Channel-to-routing coupling	Use current pipeline: scenario → channel model → link estimates → rate model → policy mask/features. Mark non-implemented models.	This is the physics-to-policy contribution.
F6	KEEP/UPDATE	Single-route evaluation flow	Load config → reset → policy/baseline → feasibility → update → terminate → aggregate seeds.	Update once corrected baselines are regenerated.
F7	KEEP/UPDATE	GNN-PPO architecture	Graph features → 3 message-passing layers → candidate scoring → masked actor; critic branch. Show LSTM only if comparator is actually evaluated.	Do not visually imply attention benefit unless ablated.
F8	NEW P0	Physics validation panel	Fiber loss vs distance; detection vs loss; QBER vs noise; rate proxy vs QBER; FSO availability/correlation.	Use analytical/limiting-case checks, not decorative curves.
F9	NEW P0	Policy physical sensitivity	Scatter/curve of P(select B) vs ΔR, plus matched QBER/outage panels.	Directly addresses the current zero-FSO-selection diagnostic.
F10	NEW P0/P1	Robustness matrix	Rows = test condition; columns = success, hops, QBER validity, FSO usage.	Use fixed seeds and report distributions.
F11	NEW P1	Training stability	Five-seed learning curves with mean + spread; optionally individual faint traces.	Demonstrate whether behavior is reproducible.
F12	NEW P1	Inference scaling	p50/p95/p99 latency vs graph size; CPU/GPU comparison if relevant.	Separate model-forward time from route-search/pre processing.
F13	NEW P2	Demand/QKP state machine	Demand → route → consume → refill → next interval.	Central figure for journal extension.
F14	NEW P2	Blocking vs traffic scaling	Traffic scaling factor on x-axis; blocking probability on y-axis; methods as series.	This is the closest direct network-level comparison to Johann.
F15	NEW P2	QKP utilization	Time-series and/or distribution of QKP filling levels; optionally regional focus.	Shows load balancing and depletion/recovery.

Fig.	Priority	Figure	How to generate it	Message/caption
F16	NEW P2	Management traffic	Controller requests/bytes vs traffic factor and method.	Must define what constitutes a management request first.

10. TABLE PLAN — WHAT SHOULD BE IN THE PAPER

Table	Purpose	Columns / content	Status
T1	Link-model provenance	Parameter, value, unit, source/evidence class, calibrated vs assumed.	**DONE conceptually; update provenance.**
T2	Observation/action features	Node features, edge features, action mask conditions.	**DONE.**
T3	Archived evaluation scenario	Endpoint, season/time, training budget, hidden size, seeds, QBER threshold, outage probability.	**DONE; replace “10 seeds” wording if it actually means evaluation seeds only.**
T4	Baseline definition	Method, information available, feasibility rule, objective, computational role.	**TODO; essential after corrected baselines.**
T5	P0 result summary	Success, hops, route length, rate utility, invalid actions, revisits, FSO usage across methods.	**TODO.**
T6	Ablation summary	Architecture/features/reward variants and metric deltas.	**P1.**
T7	Inference benchmark	Parameters, memory, p50/p95/p99, CPU/GPU, graph size.	**P1.**
T8	Journal network KPIs	Blocking, served demand, QKP utilization, depletion events, management requests.	**P2.**

11. PAPER WRITING MAP — WHERE EACH RESULT GOES

Paper section	Write this	Use these existing sources/evidence	Add before finalizing
I. Introduction	Problem: trusted-node routing must jointly account for link quality, changing channel state, trust constraints and stored-key availability. State that the current contribution is a route-level hybrid fiber/FSO controller, not a complete network-service study.	Current draft pp.1–2; Johann [4].	Make contribution claims match actual P0 evidence.
II-A. QKD background	BB84, trusted-node relay, STN distinction, and finite-key boundary.	Current draft p.2; refs [11]–[17].	Keep security claims strictly scoped.
II-B. Link/rate model	Fiber/FSO assumptions, QBER proxy, normalized rate proxy. Equation (1) exactly from current draft.	Current draft p.4; PHYSICS_EVIDENCE.md/code for implementation details; refs [18],[19] for FSO context.	Add analytical validation results.
III-A. Architecture/topology	Four simulator components, 7 hubs, topology counts, proxy limitations.	Current draft p.3; OSM [30].	Updated topology figure.
III-B. Environment	Observation, action mask, QBER threshold, route history, pool abstraction.	Current draft p.4.	Correct any feature descriptions after code audit.
IV-A. Routing formulation	Graph/POMDP notation and discounted objective Eq. (2).	Current draft p.5; refs [25],[26].	Clarify exactly what is partially observed.
IV-B. GNN/action mask	Message passing, masked policy Eq. (3), architecture figure.	Current draft p.5; refs [9],[10],[27],[28].	Add ablation evidence before claiming architectural benefit.
IV-C. PPO	Clipped objective Eq. (4); GAE/training details.	Current draft p.5; refs [8],[24].	Report protected hyperparameters and seed protocol.
IV-D. Classical comparators	BFS/Dijkstra/Max-SKR. Explain why Johann k-shortest/QKP algorithms are not apples-to-apples yet.	Current draft p.5; Johann [4]; Dijkstra [20].	Regenerate corrected baseline results.
V. Results	P0 physics validation → baseline route results → policy mechanism → robustness/generalization → inference.	Archived current results + new P0/P1 outputs.	Do not use old invalid baseline rows.
VI. Discussion	Explain what the policy learned, what it failed to learn, information/computation tradeoffs, and simulator limitations.	Controlled link-choice diagnostic + validation evidence.	Use measured language; no superiority claim from one condition.
VII. Journal extension/future work	Absolute QKP, dynamic demand, refill, blocking, load balancing, management traffic, prediction and ILP.	Johann pp.384–390; current draft limitations.	Only claim implementation after actually building it.

12. BASE-PAPER GAP MAP — WHAT TO IMPORT, WHAT NOT TO IMPORT

Johann component	Current state	Decision	When to implement
Dynamic demand matrices	**Absent**	IMPORT	P2 journal layer.
10,000 × 256-bit initial QKPs	**Absent**	IMPORT AS REPRODUCTION OPTION	Only with explicit citation/assumption; do not present as measured hardware state.
Key consumption on every path QKP	**Absent**	IMPORT	P2.
60-s refill/generation loop	**Absent**	IMPORT/ADAPT	Use exact interval only if reproducing Johann; otherwise define a justified interval.
Blocking probability	**Absent**	IMPORT	P2.
QKP load balancing/utilization	**Absent**	IMPORT	P2.
Management traffic	**Absent**	IMPORT CONCEPT	Define your own controller/query semantics before counting.
K=5 loopless k-shortest paths	**Absent**	OPTIONAL IMPORT	Useful for demand-level classica comparator; not necessary for immediate route-level paper.
Perfect prediction	**Absent**	IMPORT LATER	Only after demand history exists
LSTM prediction	**Current LSTM is graph-blind node-array encoder, not demand predictor**	DO NOT CONFUSE	Replace/augment with temporal-demand LSTM in P2 if needed.
Weight optimization	**Current draft gives the heuristic definition but does not measure it**	IMPORT LATER	Needs absolute QKP state and demand loop.
ILP	**Current draft has Eqs. (5)–(8), no measured result**	KEEP AS FORMULATION	Implement only in P2 after resource/demand layer.

Important distinction: Johann's LSTM predicts future traffic demand. The current manuscript's LSTMActorCritic is a graph-blind encoder over an ordered node-feature array. They are different objects and must not be described as the same baseline.

13. AGENT EXECUTION PROTOCOL — HOW TO WORK WITHOUT LOSING THE SCIENTIFIC THREAD

- Step 1 — Freeze provenance. Every run gets a versioned config, code commit/hash, random seed, environment version, topology version, and output directory.
- Step 2 — Never overwrite archived evidence. Keep the 25.6-hop run and 384-pair diagnostic as archived artifacts; label them preliminary.
- Step 3 — Fix the environment before comparing methods. The selector must use the observation action ordering and valid-action mask. No environment fallback should silently replace a baseline action.
- Step 4 — Separate training from evaluation. Hyperparameters are chosen on training/validation scenarios; final test seeds and held-out conditions are not used for tuning.
- Step 5 — Separate model error from policy error. First validate attenuation/QBER/rate/FSO statistics. Then test whether the policy responds correctly to those quantities.
- Step 6 — One scientific question per experiment. Do not change endpoint, weather, reward, topology and model architecture simultaneously if the goal is causal interpretation.
- Step 7 — Every plot gets a data provenance sentence. State whether points are measured, analytically calculated, simulated from assumptions, or derived from an archived policy run.
- Step 8 — Every new equation gets a provenance tag. Current draft equation, adopted literature equation, or newly derived method equation.
- Step 9 — Do not manufacture a missing comparison. If Gymnasium/runtime/dependency prevents regeneration, report the blocker and keep the old result out of the comparative table.
- Step 10 — Update the paper only after the result artifact exists. The writing agent should not draft a result claim from an intended experiment.

Recommended output directory

```
research_runs/
├── 00_archived/
├── 01_baseline_corrected/
├── 02_physics_validation/
├── 03_link_choice/
├── 04_qber_fso_stress/
├── 05_multiseed/
├── 06_generalization/
├── 07_ablations/
├── 08_inference/
└── 09_journal_network_layer/
```

Each directory should contain: config.yaml, run_manifest.json, raw_metrics.csv, summary.csv, figures/, and a short README stating the hypothesis, controlled variables, result, and paper section.

14. CLAIMS CONTROL — WHAT THE PAPER MAY SAY

Evidence state	Allowed wording	Avoid
Current simulator model	“We model…” / “The simulator computes…”	“Measured…” / “validated experimentally…”
80-km fiber / 10-km FSO outputs	“Under the configured parameters…”	“The link achieves…”
10/10 GNN archived run	“In the archived Mumbai–Kolkata condition, the policy completed 10/10 episodes.”	“The method is robust/optimal.”
25.6 hops	“Mean 25.6 hops in the archived run; the independently checked topology minimum is 24.”	“Near-optimal” unless supported by a proper route-quality study.
384-pair link diagnostic	“The policy selected the higher-proxy FSO candidate in 0/384 controlled comparisons.”	“The GNN understands/does not understand physics” — too broad.
P0 multi-seed result	“Across five independent training seeds…”	Single-run generalization claims.
Journal blocking result	“Under the specified demand/QKP model…”	General claims about deployed QKD network throughput.
ILP	“We formulate an ILP comparator…”	“ILP outperforms the learned policy” before execution.
Field relevance	“Provides a simulation framework for…”	“Demonstrates real-world network performance.”

15. MASTER PRIORITY QUEUE

Priority	Do this	Definition of done	Paper impact
P0.1	Regenerate corrected baselines	All methods use same mask, endpoint, seeds, termination and metrics; raw CSV archived.	Unlocks valid Results comparison.
P0.2	Validate physics/noise	Analytical trends + independent calculations + FSO stochastic checks pass.	Makes channel model defensible.
P0.3	Run controlled link-choice/QBER/FSO tests	Policy probability and selection behavior quantified against controlled physical changes.	Explains mechanism/current weakness.
P0.4	Run 5 independent training seeds	Fixed held-out evaluation; mean/spread reported.	Removes single-run limitation
P1.1	Endpoint/season/diurnal generalization	Held-out conditions complete with confidence intervals.	Makes robustness claim possible.
P1.2	Feature/architecture/reward ablations	Same training budget and seed distribution; effects quantified.	Explains what drives policy.
P1.3	Inference benchmark	p50/p95/p99, memory, CPU/GPU and scaling reported.	Adds implementation/deployment relevance.
P1.4	Small-graph oracle	Exact feasible-path reference and regret computed.	Adds route-quality reference.
P2.1	Absolute QKP model	Capacity, initial inventory, generation and consumption units defined.	Unlocks network-service metrics.
P2.2	Dynamic demand loop	Demand → route → consume → refill works over time.	Turns route study into network study.
P2.3	Blocking/load balancing	Traffic scaling + QKP utilization + recovery measured.	Direct Johann comparison.
P2.4	Management traffic	Controller query semantics and overhead measured.	Information/security tradeoff
P2.5	Weight/LSTM/perfect prediction	Classical global-state and temporal-demand comparators implemented.	Journal baseline family.
P2.6	ILP	Demand-level capacity model solved at defined scale; runtime recorded.	Global reference/oracle.
P3	Measured traces / HIL	Trace provenance + model-vs-measurement error + policy transfer reported.	External validation.

Recommended immediate stopping point: finish P0 and P1 before building the full Johann-style demand/QKP layer. The immediate paper then has a complete scientific story: noise-aware hybrid network model → constrained routing problem → GNN-PPO controller → corrected classical comparison → policy mechanism → robustness/generalization → inference cost → explicit limitations.

Journal story after P2: extend that same controller into a sustained network-service environment where routing decisions consume and replenish finite QKD resources under dynamic demand, enabling blocking, load balancing, management overhead, prediction, and global optimization comparisons.

16. SOURCE LOCATOR — QUICK REFERENCE FOR THE AGENT

Source	Where	Use
Current QKD_IEEE_Paper.pdf	p. 1	Abstract, contribution boundary, current evidence scope.
Current QKD_IEEE_Paper.pdf	p. 2	QKD/TN/STN background, finite-key boundary, rate interpretation.
Current QKD_IEEE_Paper.pdf	p. 3	Architecture, topology, parameter table, proxy-network assumptions.
Current QKD_IEEE_Paper.pdf	p. 4	FSO model, Eq. (1), observation features, action mask, archived setup.
Current QKD_IEEE_Paper.pdf	p. 5	Simulation flow, Eq. (2)–(4), GNN/action mask, PPO, weight heuristic.
Current QKD_IEEE_Paper.pdf	p. 6	ILP Eqs. (5)–(8), preliminary 10/10 and 25.6-hop result, key-store limitation.
Current QKD_IEEE_Paper.pdf	p. 7	Physics evidence, controlled link-choice diagnostic, scope/validation limitations.
Current QKD_IEEE_Paper.pdf	p. 8	References [14]–[30].
Johann et al. JOCN 2024	p. 384	Absolute QKD key-generation model: QBER, post-processing factor, secret-key block calculation.
Johann et al. JOCN 2024	p. 385	Central controller, QKP architecture, security/management-information tradeoff.
Johann et al. JOCN 2024	p. 386	Dynamic demand matrix processing, key consumption, refill, K=5 loopless shortest paths.
Johann et al. JOCN 2024	p. 387	Perfect prediction and LSTM demand prediction.
Johann et al. JOCN 2024	p. 389	Blocking probability and QKP utilization/load balancing.
Johann et al. JOCN 2024	p. 390	Management traffic and overall tradeoffs.
Uploaded QSMS Presentation Script	pp. 1–11	Visual reference only: DejaVu Sans, burgundy #9E1735, pale #F6F1F2, white background, restrained tables and sectioning

This roadmap is designed so another research agent can resume work without reconstructing the project history: start at Section 2 for current state, Section 3 for missing dependencies, Section 4 for P0, Section 7 for equation provenance, Section 9 for figure recipes, and Section 11 for exact paper placement.

RESEARCH ROADMAP

STATUS UPDATE - 2026-10-07

How to read this update. This addendum supplements the original 22-page roadmap. It records work completed in the repository, generated evidence, and remaining dependencies. Earlier archived results remain historical records and are not silently replaced. All current outputs are simulator-derived; none is field validation or a QKD security proof.

1. Work completed in this update

Workstream	Change / result	Remarks and evidence
Corrected route comparison	Baseline selectors use the observation's feasible-action ordering; all methods use the same season and channel seeds. Per-episode output records action validity, no-feasible-action states, fallbacks, route, hops, revisits, and link validity.	The comparison output is identified below. No-feasible-action states are separated from invalid policy actions.
Physics consistency	Independent formula transcription covered 201 fiber distances from 0 to 200 km. Maximum absolute difference: QBER 0; normalized proxy 0. Transmission and proxy decrease; QBER does not decrease.	Internal consistency only. It does not validate assumptions against measurements or establish security.
FSO stochastic checks	All 16 configured season/distance anchors sampled at 2,000 trials each; maximum absolute availability deviation 0.0125 (1.22 target standard errors). Log-Cn2 AR(1): configured 0.85, sampled 0.8522 over 20,000 steps.	These are model targets and samples, not validation against Indian weather observations.
Link-feature sensitivity	The controlled diagnostic includes a proxy-rate ratio sweep, a feasible QBER-margin sweep, and an outage-mask counterfactual with route context held fixed. The FSO action probability rose from 0.311 to 0.638 as its proxy-rate ratio rose from 0.25x to 4x; at QBER 0.111 it was masked. In a separate 32-sample FSO turbulence counterfactual, a 100x Cn2 multiplier lowered median proxy from 0.202 to 0.183 and mean FSO action probability from 0.749 to 0.740; FSO remained selected in 100% of samples.	A policy-input diagnostic does not establish link-measurement validity or route-level performance.
Independent training-seed stability	Five independent checkpoints per architecture were completed. GNN mean success 46.1% (SD 16.8%; nested 95% interval 30.9%-58.5%); MLP mean success 79.8% (SD 9.4%; nested interval 71.3%-87.4%).	Variability is across fitted policies and scenario clusters in this simulator protocol; it is not external validation.
Exact toy-graph route oracle	Exhaustive simple-path reward search covered 20 graph/channel cases with a feasible oracle in 20. The policy succeeded in 90%; it exactly matched oracle reward in 55% of all cases and 61% of successful cases.	Five generated 7-node graphs; episode link states frozen with 5% per-edge outage. This exact result is limited to the toy static-channel protocol, not the full dynamic graph.
Paired endpoint / season / time transfer	Across 108 GNN episodes, success was 99/108 (91.7%; Wilson 95% CI 84.9%-95.6%); successful mean hops 31.94. BFS-hop success was 108/108 (100.0%), with 15.40 successful mean hops. Paired exact McNemar p=0.0039; among jointly successful episodes, GNN-minus-BFS hop difference was 16.00 (95% pair-cluster bootstrap CI 6.48 to 25.97). GNN did not outperform BFS.	Six ordered pairs, four seasons and selected day/night times, paired seeds. BC endpoint randomization means these are not strict end-to-end endpoint holdouts. Hop interval resamples only six endpoint-pair clusters.
Repeated inference profile	1 thread(s): p50/p95/p99 26.52/70.37/94.68 ms; 4 thread(s): p50/p95/p99 22.13/45.12/64.58 ms; 8 thread(s): p50/p95/p99 12.76/16.89/22.58 ms. Process RSS after run 348.4 MiB; model parameters 0.5 MiB.	42 ordered pairs, deterministic single-decision calls, encoder cache disabled. One host/topology; graph-size scaling and deployment profiling remain open.
Matched-budget ablation campaign	0/5 conditions complete at the shared 50-epoch budget. Current condition: baseline, ppo phase, PPO 25/50.	Single training seed; progress is not a completed comparative result.
Strict matched held-out evaluation	Five GNN and five MLP policies were evaluated on the same 108 held-out scenarios. Training-seed mean success: GNN 46.1% (SD 16.8%; nested 95% interval 30.9%-58.5%), MLP 79.8% (SD 9.4%; nested interval 71.3%-87.4%), BFS 100.0%. Mean successful hops across seeds: GNN 87.77, MLP 57.80. At 10% fiber outage, GNN/MLP mean success was 46.3%/80.6%.	Paired scenario IDs; five fitted policies per learned architecture; intervals resample training seeds outside scenario clusters. Simulator results, not field validation.

Workstream	Change / result	Remarks and evidence
Final-checkpoint reward decomposition	One final GNN seed and its matched MLP seed were decomposed over 108 scenarios against BFS-hop. Route success: GNN 46.3%, MLP 87.0%, BFS 100.0%.	Reward components are descriptive shaped-return contributions; route metrics remain the primary comparison and this one-seed audit does not estimate seed uncertainty.

2. Paired route comparison

Policy	Success	Successful mean hops	Invalid actions	No-valid-action states	Revisits
GNN-PPO	16/16 (100.0%; 95% CI 80.6%-100.0%)	28.25	0	0	2
Random	0/16 (0.0%; 95% CI 0.0%-19.4%)	n/a	0	8	2459
Dijkstra-km	4/16 (25.0%; 95% CI 10.2%-49.5%)	233.00	0	0	172
BFS-hop	16/16 (100.0%; 95% CI 80.6%-100.0%)	28.25	0	0	3
Max-SKR	0/16 (0.0%; 95% CI 0.0%-19.4%)	n/a	0	5	984

Conditions: Delhi to Chennai; 4 shared environment seeds per each of 4 seasons; QBER/rate action mask; 400 step limit. Scope is single-route simulator behavior, not traffic blocking, network-wide key-pool balance, or delivered service capacity.

Checkpoint transfer across city pairs

The same 50-epoch checkpoint completed 31/42 ordered city-pair episodes (73.8%) under the archived monsoon-night scenario. Median successful route length was 20 hops; localized decision latency was 3.83 ms median and 4.34 ms p95 on this host. This is one stochastic sample per ordered pair, not a held-out endpoint protocol or a matched baseline comparison.

3. Roadmap status after this work

Priority	Status	Evidence / remaining work
P0-1 corrected route comparison	DONE - route-level	Paired environment seeds, shared feasibility rules, and action/fallback accounting. The separate demand-driven QKP comparison remains out of scope.
P0-2 physics checks	DONE - internal only	Fiber formula consistency, configured FSO anchor sampling, and temporal correlation. No matched device/channel calibration or measured traces.
P0-3 link sensitivity	DONE - diagnostic	Rate-ratio, QBER-margin, and hard outage-mask tests vary simulator policy inputs only.
P0-4 QBER stress	PARTIAL	Controlled action-probability sweep covers QBER 0.01, 0.03, 0.05, 0.07, 0.09, 0.10, 0.109 and 0.111. The out-of-threshold candidate is masked. Route success, entropy, and preference curves across endpoint/weather conditions remain.
P0-5 FSO / outage stress	DONE - paired route stress	Paired route outcomes are archived at outage probabilities 0%, 1%, 5%, 10%. These are simulator stress scenarios; pointing/background parameters remain fixed and outage-persistence bursts are not separately calibrated.
P0-6 five training seeds	DONE - five GNN + five MLP	Five GNN and five matched MLP full-budget runs are complete and their checkpoint hashes are recorded in the strict evaluation report.
P1 endpoint / weather holdouts	DONE - strict paired test	Final report covers 108 paired held-out endpoint/condition/scenario IDs, five trained seeds each for GNN and MLP, four fiber-outage levels, and per-checkpoint CPU latency. See the final report and manuscript for estimates and uncertainty.
P1 architecture / feature ablations	DEFERRED - outside current scope	An earlier campaign stopped during the baseline at PPO epoch 25/50; no ablation comparison is used in the manuscript and the campaign is deferred.
P1 inference cost	DONE - one-host protocol	Per-decision latency was measured for all ten final checkpoints with warm-up, four CPU threads, and caching disabled. Results characterize one host/topology only; graph-size scaling and deployment profiling remain open.
P1 exact small-graph oracle	DONE - toy scope	Exhaustive simple-path reward oracle and policy regret archived for generated 7-node graphs with frozen episode channels. Does not establish optimality on the full dynamic network.
P2 network-service study	BLOCKED BY MODEL SCOPE	Requires absolute key generation and QKP capacity, time-indexed demand, consumption/refill, blocking, load balance, controller message semantics, prediction baselines, and executable demand-level ILP.
P3 external validation	BLOCKED BY DATA / HARDWARE	Requires matched device/channel measurements or hardware-in-the-loop and separate model-error analysis.

4. Interpretation and next work

The corrected comparison supports a route-level simulator statement only. Report success, successful-route hops, revisits, action-mask violations, and route outcomes; do not infer network capacity or superiority from reward alone. Preserve baseline failures as observed outcomes and state each baseline definition.

The strict five-seed endpoint/weather and paired outage evaluation, selected-checkpoint reward decomposition, final figures, and revised manuscript are complete. Before submission, the authors should fill the author/funding block and verify the final package in their target venue template. The demand/QKP service-capacity study and external device/weather validation remain deferred extensions outside this completion. Do not present the demand-level BB84/QKP formulation as implemented until an absolute key-resource and demand model is adopted and validated.

Regression check: 41 tests passed, 0 failed in 68.8 s; 1 warning(s) recorded. Details: `test_results_20261002.json`.

Provenance rule: Label every numeric result as measured, independently calculated, model-sampled, or derived from a policy run. Internal consistency checks do not upgrade an assumption to calibrated physics.

RESEARCH ROADMAP

MATCHED GNN-PPO / LSTM-PPO UPDATE - 2026-10-08

The revised manuscript now compares only GNN-PPO with the repository's matched LSTM-PPO node-sequence encoder. Five checkpoints per architecture were evaluated on the same 108 scenario IDs per checkpoint (540 episodes per method). The table reports simulator results with uncertainty across fitted checkpoints and shared scenario clusters; it is not field validation.

Measure	GNN-PPO	LSTM-PPO / paired result
Route completion (GNN-PPO)	249/540 (46.1%); nested 95% CI 31.3%-58.5%	Per-seed range 18.5%-63.9%; successful-hop checkpoint mean 87.77
Route completion (LSTM-PPO)	403/540 (74.6%); nested 95% CI 58.0%-87.6%	Per-seed range 42.6%-89.8%; successful-hop checkpoint mean 71.39
Paired completion effect		LSTM minus GNN +28.5%; nested 95% CI +20.6% to +37.2%
Matched outcomes		Both succeeded 242; GNN-only success 7; LSTM-only success 161; both failed 130
Successful-route hops		Among 242 episodes where both succeeded, mean LSTM-minus-GNN difference -32.56 hops
CPU actor-call p95 (GNN-PPO)	9.610 ms	Across-checkpoint SD 1.647 ms
CPU actor-call p95 (LSTM-PPO)	6.241 ms	Across-checkpoint SD 0.216 ms

Interpretation and remaining scope

The paired bootstrap is the primary uncertainty summary. Successful-hop differences are conditional on both methods completing the same scenario. The LSTM reads node-feature rows in a fixed topology order, has no graph-edge input, and resets its state on every action call; the comparison is therefore between these two implementations, not graph memory versus persistent temporal memory. Reward-component totals are descriptive because trajectories can differ in length. The manuscript figures are constrained to the sections that interpret them, and the publication bundle excludes older route-baseline and GNN/MLP comparisons retained in the local research archive.

The study is simulator-only. Field channel calibration, absolute key-pool and multi-demand service modeling, network blocking/capacity, broader topology tests, and a formal composable-security proof remain outside the completed paper revision.

The selected-seed reward audit is complete on 108 matched scenarios. Its cumulative reward-component report and episode ledger are included in the supplementary package; the plot is a descriptive diagnostic only.